

Maximum Likelihood Estimation for Locally Stationary Hidden Markov Models

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Structure

- What is a Hidden Markov model?
- What does *locally stationary* mean?
- Adaptation of a classical consistency proof.

What is a Hidden Markov Model?

An introductory example

- Imagine **A**lice and **B**ob use a **C**hannel for signal transmission.
- **A** sends signals $(X_0, \dots, X_{n-1})$ where X_k takes values in $\{0, 1\}$.

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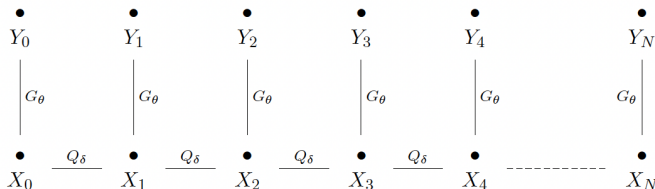
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- The array of signals will be assumed to be a Markov chain, i. e.

$$P(X_k = i | X_{k-1}, \dots, X_0) = P(X_k = i | X_{k-1})$$
- Transition probabilities $Q_\delta(j, \{i\}) = P_\delta(X_k = i | X_{k-1} = j)$ are known.

Situation can be mathematically modeled in the following way.

- Consider $(\mathcal{X} := \{0, 1\}, \mathcal{B}(\mathcal{X}))$ and $(\mathcal{Y} := \{0, 1\}, \mathcal{B}(\mathcal{Y}))$.
- (X_k, Y_k) is a bivariate Markov chain on $\mathcal{X} \times \mathcal{Y}$ with the following dependence structure.



Say the kernels are represented by the matrices

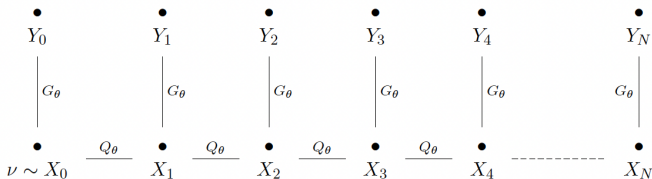
$$Q_\delta = \begin{pmatrix} Q_\delta(0,0) & Q_\delta(1,0) \\ Q_\delta(0,1) & Q_\delta(1,1) \end{pmatrix} = \begin{pmatrix} 1-\delta & \delta \\ \delta & 1-\delta \end{pmatrix} \text{ for } \delta \in (0,1)$$

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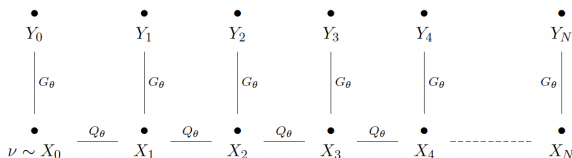
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- **B** knows δ and could now resort to the highly developed theory of *Hidden Markov models* – for example Maximum likelihood estimation for $\theta \in [\delta, 1-\delta]$.
- Hidden Markov Models are models of the above form. In the *classical and well-understood* homogeneous case



The formal definition

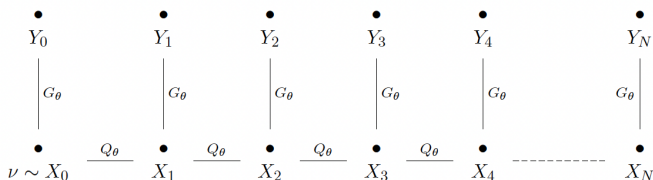


Definition (Hidden Markov Models)

Let $(\mathcal{X}, \mathcal{A}_\mathcal{X})$ and $(\mathcal{Y}, \mathcal{A}_\mathcal{Y})$ be measurable spaces and let $(Z_k)_k = ((X_k, Y_k))_k$ be a bivariate Markov chain on $\mathcal{X} \times \mathcal{Y}$. We say that $(Z_k)_k$ is a Hidden Markov Model, if for every k there is a Markov kernel Q_k from $(\mathcal{X}, \mathcal{A}_\mathcal{X})$ to itself and a Markov kernel G_k from $(\mathcal{X}, \mathcal{A}_\mathcal{X})$ to $(\mathcal{Y}, \mathcal{A}_\mathcal{Y})$ such that for every bounded and measurable $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ we have

$$\mathbb{E} [f(X_k, Y_k) | X_0^{k-1}, Y_0^{k-1}] = \int_{\mathcal{X}} \int_{\mathcal{Y}} f(x, y) G_k(x, dy) Q_k(X_{k-1}, dx)$$

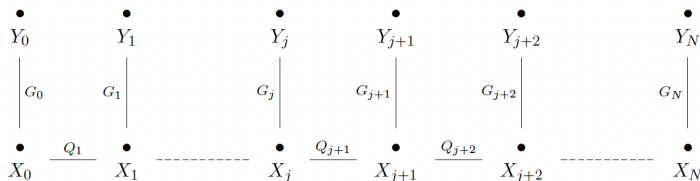
Applications of Hidden Markov Models



- Real-world phenomena X_k that are only observed via noisy or incomplete data.
 - physical quantities observed in experiments
 - evolution of medical indicators whose measurements are noisy
 - ...
- live-tracking of moving objects
- ...

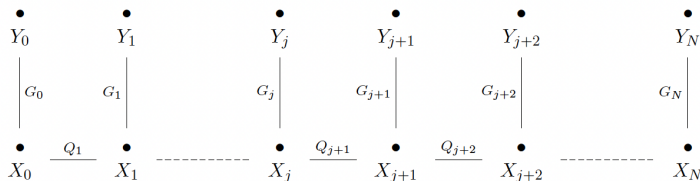
What is local stationarity?

The issue of homogeneity and stationarity



- Homogeneity/Stationarity is a key requirement for the general consistency results on maximum likelihood estimators to hold.

The issue of homogeneity and stationarity



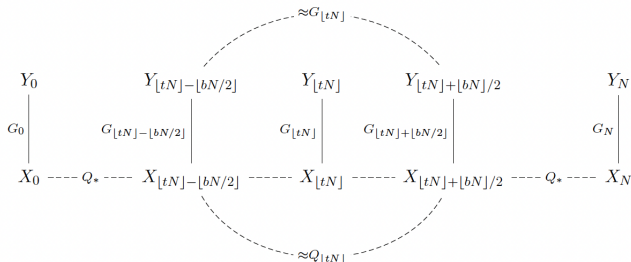
- Homogeneity/Stationarity is a key requirement for the general consistency results on maximum likelihood estimators to hold.
- ... and also to somewhat even make sense.
- $\hat{\theta}_N \in \operatorname{argmax}_{\theta \in \Theta} \log \rho_{\theta}(Y_0^N)$ not expected to approximate anything useful, if transition parameters evolve *drastically* over time.

Typical situation:

- Provided with an array of $N + 1$ many observations $Y_0, \dots, Y_N$
- and reasonable to assume transition kernels Q and G are approximately the same around $\lfloor tN \rfloor$ for $\approx bN$ many observations.
- Aim to find out what the kernels around $\lfloor tN \rfloor$ look like.

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- Pretend homogeneity around $\lfloor tN \rfloor$
- and do maximum likelihood estimation *anyway* only including the observations from that window of size $\approx bN$

In conclusion, consider a maximum likelihood estimator resembling

$$\hat{\theta}_N \in \operatorname{argmax}_{\theta \in \Theta} \log \rho_{\theta}(Y_{\lfloor tN \rfloor - \lfloor bN/2 \rfloor}^{\lfloor tN \rfloor + \lfloor bN/2 \rfloor})$$

and hope $Q_{\hat{\theta}_N} \approx Q_k$ and $G_{\hat{\theta}_N} \approx G_k$ for

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- Need an **asymptotic theory** for this estimator (and for other methods of statistical inference as well).
- In our situation, including more data from $Y_0, \dots, Y_N$ is **not an option**.
- I. e. no classical asymptotic theory for $\hat{\theta}_n \in \operatorname{argmax}_{\theta \in \Theta} \log \rho_{\theta}(Y_0^n)$, $n \rightarrow \infty$.

Developing an adequate asymptotic

$$\hat{\theta}_N \in \operatorname{argmax}_{\theta \in \Theta} \log \rho_{\theta}(Y_{\lfloor tN \rfloor - \lfloor bN/2 \rfloor}^{\lfloor tN \rfloor + \lfloor bN/2 \rfloor})$$

- Effectively, we estimate a parameter at a point $t \in [0, 1]$ in *unit time*.
- Good would be an asymptotic $b \rightarrow 0$ but still $bN \rightarrow \infty$.
- We get **both** more stationary around every $t \in [0, 1]$ and gather more observations as $N, bN \rightarrow \infty$

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$$\|P - Q\|_{TV} = \sup_{A \in \mathcal{A}} |P(A) - Q(A)|.$$

Definition (First (heuristic) definition for local stationarity)

- A triangular scheme $\{\{Z_{k,N}\}_{0 \leq k \leq N}\}_{N \geq 0}$ of processes with finite dimensional distributions $\pi_{k,N}^{m,N}$
- And for every $t \in [0, 1]$ a stationary process $(\tilde{Z}_k(t))_{k \in \mathbb{Z}}$ with finite dimensional distributions $\pi_k^m(t) = \pi_{m-k+1}(t)$
- $\limsup_{N \rightarrow \infty} \sup_{0 \leq k \leq m \leq N, |m-k| \leq l} \|\pi_k^m(k/N) - \pi_{k,N}^{m,N}\|_{TV} = 0 \quad \forall l \geq 1$

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Of course, variations possible

- Other metric on the distributions' space possible
- And definition involves subtleties (continuity of limit distributions)
- Often possible to derive explicit bounds of the form

$$\|\pi_k^m(u) - \pi_{k,N}^{m,N}\|_{TV} \leq C_l \left[\left| u - \frac{k}{N} \right| + \frac{1}{N} \right] \text{ for } 0 \leq m - k \leq l$$

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What does the MLE look like now? Morally at least like

$$\hat{\theta}_N(t) \in \operatorname{argmax}_{\theta \in \Theta} \log \rho_{\theta}(Y_{[tN] - \lfloor bN/2 \rfloor, N}^{[tN] + \lfloor bN/2 \rfloor, N})$$

What happens as $N \rightarrow \infty$, $b \rightarrow 0$ but yet still $bN \rightarrow \infty$?

How can we adapt a consistency proof?

The classical case

First let us go back to **assuming stationarity**.

- In the classical iid case we decompose

$$n^{-1} \log \rho_{\theta}(X_0^{n-1}) = n^{-1} \sum_{k=0}^{n-1} \log \rho_{\theta}(X_k) \xrightarrow{n \rightarrow \infty} E [\log \rho_{\theta}(X_0)] =: \ell(\theta)$$

by the LLN. Then use separation/identifiability properties of $\ell(\theta)$ (unique maximum at θ^*) and – for example – uniformity of the convergence.

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- In Hidden Markov case it is harder.

$$\begin{aligned} n^{-1} \log \rho_{\theta}(Y_0^{n-1}) &= n^{-1} \sum_{k=0}^{n-1} \log \left(\rho_{\theta}(Y_0^k) / \rho_{\theta}(Y_0^{k-1}) \right) \\ &= n^{-1} \sum_{k=0}^{n-1} \log \rho_{\theta}(Y_k | Y_0^{k-1}) \end{aligned}$$

The likelihood function's (slightly) nasty explicit form

Explicit form:

$$\log \rho_{\theta}(Y_k | Y_0^{k-1}) = \log \int_{\mathcal{X}} \gamma_{\theta}(x_k, Y_k) Q_{\theta}(x_{k-1}, dx_k) P_{\theta}(X_{k-1} \in dx_{k-1} | Y_0^{k-1})$$

γ_{θ} transition density, i.e.

$$\int_{\mathcal{X}} \int_{\mathcal{Y}} \mathbb{1}_C(x', y') \gamma_{\theta}(x', y') \phi(dy') Q_{\theta}(x, dx') = \int_{\mathcal{X}} \int_{\mathcal{Y}} \mathbb{1}_C(x', y') G_{\theta}(x', dy') Q_{\theta}(x, dx')$$

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- Say $\gamma_{\theta}(x, y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-y)^2}{2\sigma^2})$ and $\mathcal{X}$ is a finite state space $\mathcal{X} = \{\mu_1, \dots, \mu_d\}$.
- ... or say $\mathcal{X}$ is a compact manifold of dimension k , Q has diffusion type transition density $q^{(t_0)}(x, x') \sim (4\pi t_0)^{-k/2} e^{-d(x, x')^2/(4t_0)}$
- ... or try to figure out densities in noisy channel example.

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⇒ **We don't do this!**

What one usually does in the Hidden Markov case

$$n^{-1} \log \rho_{\theta}(Y_0^{n-1}) = n^{-1} \sum_{k=0}^{n-1} \underbrace{\log \rho_{\theta}(Y_k | Y_0^{k-1})}_{g_k(Y_0^k)}.$$

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- Remember: $n^{-1} \sum_{k=0}^{n-1} g(Y_{k-m+1}, \dots, Y_k) \rightarrow E[g(Y_0, \dots, Y_{m-1})]$ almost surely,
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- Have to always include the same amount of m observations (approximation depth).
- There is a term $\ell(\theta)$ independent of the approximation depth (corresponding to $m = \infty$). Show now:

$$n^{-1} \sum_{k=0}^{n-1} \log \rho_{\theta}(Y_k | Y_{k-m+1}^{k-1}) \rightarrow \ell(\theta) + \varepsilon.$$

What we need to do in the **locally stationary** case

$$n^{-1} \sum_{k=0}^{n-1} \log \rho_{\theta}(Y_k | Y_{k-m+1}^{k-1}) \longrightarrow \ell(\theta) + \varepsilon$$

- If homogeneity (stationarity) fails, ergodic theorem makes no sense.
- Especially in our triangular array setting.

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- If homogeneity (stationarity) fails, ergodic theorem makes no sense.
- Especially in our triangular array setting.
- Need a suitable LLN for triangular arrays.
- The above introduced *finite memory approximation* is in the spirit of local stationarity.

Treating with the likelihood function

$$(bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \log \rho_\theta(Y_{k,N} | Y_{t_N^-, N}^{k-1, N})$$

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$$\begin{aligned} &= (bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\log \rho_\theta(Y_{k,N} | Y_{k-l, N}^{k-1, N}) - \mathbb{E} \left[\log \rho_\theta(Y_{k,N} | Y_{k-l, N}^{k-1, N}) \right] \right) \\ &\quad + (bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\mathbb{E} \left[\log \rho_\theta(Y_{k,N} | Y_{k-l, N}^{k-1, N}) \right] - \mathbb{E} \left[\log \rho_\theta(Y_k(t) | Y_{k-l}^{k-1}(t)) \right] \right) \\ &\quad + (bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \mathbb{E} \left[\log \rho_\theta(Y_k(t) | Y_{k-l}^{k-1}(t)) \right]. \end{aligned}$$

A LLN for triangular arrays

$$(bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\log \rho_\theta(Y_{k,N} | Y_{k-1,N}^{k-1,N}) - \mathbb{E} \left[\log \rho_\theta(Y_{k,N} | Y_{k-1,N}^{k-1,N}) \right] \right)$$

Proposition

Let $\{\{Z_{k,N}\}_{0 \leq k \leq N}\}_{N \geq 0}$ be a triangular array of random variables which is adapted to the family of filtrations $\{\mathbb{F}_N = (\mathcal{F}_{k,N})_{0 \leq k \leq N}\}_{N \geq 0}$. Assume there is a C and a $1 > \rho > 0$ such that

$$|\mathbb{E}[Z_{k,N} | \mathcal{F}_{m,N}]| \leq C \cdot \rho^{k-m},$$

then for $bN \rightarrow \infty$

$$(bN)^{-1} \sum_{k=t_N^-}^{t_N^+} Z_{k,N} \xrightarrow{\mathbb{P}} 0.$$

How can we adapt a consistency proof?

$$(bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N}) - \mathbb{E} \left[\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N}) \right] \right)$$

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- Deterministic bound on $\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N})$ strong but possible.
- Geometrical decay (forgetting) non-trivial.

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Definition (Doebelin condition)

Let K be a Markov kernel from $(\mathcal{Z}_1, \mathcal{A}_{\mathcal{Z}_1})$ to $(\mathcal{Z}_2, \mathcal{A}_{\mathcal{Z}_2})$. If there is a probability measure μ on $(\mathcal{Z}_2, \mathcal{A}_{\mathcal{Z}_2})$ and an ε such that

$$K(x, A) \geq \varepsilon \mu(A) \text{ for all } A \in \mathcal{A}_{\mathcal{Z}_2},$$

we say that Q satisfies a Doeblin condition.

The quite powerful Doeblin condition

$$K(x, A) \geq \varepsilon \mu(A) \text{ for all } A \in \mathcal{A}_{\mathcal{Z}_2},$$
$$|E[Z_{k,N} | \mathcal{F}_{m,N}]| \leq C \cdot \rho^{k-m}$$

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$$|E[Z_{k,N} | \mathcal{F}_{m,N}]| \leq C \cdot \rho^{k-m}$$

Proposition (Doeblin condition leads to a contraction)

If $Q_1, \dots, Q_n$ are Markov kernels on $(\mathcal{X}, \mathcal{A}_{\mathcal{X}})$ and satisfy the Doeblin condition with common ε , we get for every $\nu \in \mathcal{M}_0(\mathcal{X}, \mathcal{A}_{\mathcal{X}})$ the bound

$$\|Q_n \cdots Q_1 \nu\|_{\text{TV}} \leq (1 - \varepsilon)^n \|\nu\|_{\text{TV}}.$$

- Think of $\nu = \delta_x - \delta_y$ for $x, y \in \mathcal{X}$.
- Doeblin $\rightsquigarrow$ exponential forgetting of initial condition $\rightsquigarrow$ convergence against expected value

The quite powerful Doeblin condition

$$K(x, A) \geq \varepsilon \mu(A) \text{ for all } A \in \mathcal{A}_{\mathcal{Z}_2},$$

$$|E[Z_{k,N} | \mathcal{F}_{m,N}]| \leq C \cdot \rho^{k-m}$$

Proposition (Doeblin condition leads to a contraction)

If $Q_1, \dots, Q_n$ are Markov kernels on $(\mathcal{X}, \mathcal{A}_{\mathcal{X}})$ and satisfy the Doeblin condition with common ε , we get for every $\nu \in \mathcal{M}_0(\mathcal{X}, \mathcal{A}_{\mathcal{X}})$ the bound

$$\|Q_n \cdots Q_1 \nu\|_{\text{TV}} \leq (1 - \varepsilon)^n \|\nu\|_{\text{TV}}.$$

- Think of $\nu = \delta_x - \delta_y$ for $x, y \in \mathcal{X}$.
- Doeblin $\rightsquigarrow$ exponential forgetting of initial condition $\rightsquigarrow$ convergence against expected value
- $(bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\log \rho_{\theta}(Y_{k,N} | Y_{k-l,N}^{k-1,N}) - E \left[\log \rho_{\theta}(Y_{k,N} | Y_{k-l,N}^{k-1,N}) \right] \right)$ converges

The (Doebelin-powered) final appearance of local stationarity

$$(bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\mathbb{E} \left[\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N}) \right] - \mathbb{E} \left[\log \rho_\theta(Y_k(t) | Y_{k-l}^{k-1}(t)) \right] \right) \left| \int f d\nu \right| \leq \|f\|_\infty \|\nu\|_{\text{TV}}$$

Theorem ($\|\cdot\|_{\text{TV}}$ -local stationarity of Markov chains, [Tru19])

Let $\{K_u\}_{u \in [0,1]}$ be a family of Markov kernels on $(\mathcal{Z}, \mathcal{B}(\mathcal{Z}))$ and let $\{\{Z_{k,N}\}_{0 \leq k \leq N}\}_{N \geq 0}$ be a triangular array of Markov chains with transition kernels $P(Z_{k,N} \in dx | Z_{k-1,N}) = K_{\frac{k}{N}}(Z_{k-1,N}, dx)$ that satisfies the following two conditions:

- (A1) All transition kernels K_u satisfy a Doeblin condition with common constant $\varepsilon > 0$.
- (A2) $\sup_{z \in \mathcal{Z}} \|K_u(z, \cdot) - K_v(z, \cdot)\|_{\text{TV}} \leq L|u - v|$

Then, for every l we have

$$\|\pi_{k,N}^{k+l,N} - \pi_{l+1}(t)\|_{\text{TV}} \leq C_{l+1} \left(\left| t - \frac{k}{N} \right| + \frac{1}{N} \right).$$

On possible assumptions

- We use $|\log \rho_\theta(Y_{k,N} | Y_0^k)| \leq C$ type bounds for total variation
 $\rightsquigarrow \kappa \leq \gamma_\theta \leq \kappa^{-1}$ for $\kappa \in (0, 1)$.

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Theorem

Assume the transition kernels $T_{k,N}$ of the array of Hidden Markov models are given by $T_{\theta^\infty(k/N)}$ for $\theta^\infty : [0, 1] \rightarrow \Theta$ Lipschitz. Assume further that the above assumptions hold. Then, the maximum likelihood estimator is (weakly) consistent.

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Exact consistency depends on identifiability – but this is a property of the stationary distributions.

The Doeblin condition and possible examples

- $\mathcal{X} = \{x_1, \dots, x_n\}, \mathcal{Y} = \{y_1, \dots, y_m\}, \inf_{\theta} \inf_{i,j \in \mathcal{X}} q_{\theta}(i,j) > 0, \kappa \leq \gamma_{\theta} \leq \kappa^{-1}.$
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- But even in simple finite state space models, Doeblin minorization is often not fulfilled: $\mathcal{X} = \{1, 2, 3\},$

$$q_{\theta}(i,j) = \begin{cases} \theta & \text{if } j = i + 1 \pmod{3} \\ 1 - \theta & \text{if } j = i - 1 \pmod{3} \\ 0 & \text{else.} \end{cases}$$

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- On the level of random variables,

$$f(x, X_{k-1}) = \begin{cases} V_{\nu} & \text{if } U \leq \varepsilon \\ g_K(X_{k-1}, W) & \text{if } U \geq \varepsilon \end{cases}$$

for $U, W \sim \mathcal{U}([0, 1])$ and $V_{\nu} \sim \nu$ independent and g_K suitable.

Potential modifications and the AR(1) case

- Gaussian observation densities γ_θ also possible in slightly different setting $\rightsquigarrow$ mixture models, $\mathcal{X}$ could encode changing variances and means.
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- The seminal paper [Dou+11] covered such AR(1) models (and many other models from classical TSA).
- Quite intricate usage of ergodic theorem, seems to intentionally circumvent invoking (stronger) forgetting properties.

All is not lost!

Definition (L^p -local stationarity in the spirit of [Dah12])

A triangular array of Hidden Markov chains $\{\{(X_{k,N}, Y_{k,N})\}_{0 \leq k \leq N}\}_{N \geq 0}$ is called L^p -locally stationary, if for every $u \in [0, 1]$ there is a stationary Hidden Markov chain $\{(\widetilde{X}_k(u), \widetilde{Y}_k(u))\}_{k \in \mathbb{Z}}$ such that for every sequence k such that $\frac{k}{N} \rightarrow u$ we have $\|\varepsilon_{k,N}(u) := Y_{k,N} - \widetilde{Y}_k(u)\|_p \rightarrow 0$.

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- The first term converges by the stationary case.
- Under the right assumptions, d vanishes in L^p by interpolation arguments.

More is possible! But be vigilant!

- Actually, in classical (Doebelin-) cases

$$\sup_{u \in [\frac{b}{2}, 1 - \frac{b}{2}]} \|\hat{\theta}_n(u) - \theta^*(u)\| \xrightarrow{P} 0.$$

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Thank you for your attention!

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